

25CS1201E | Frontend Development Frameworks & UI Engineering

Examination

Invigilation Duty Allocator

Smart Web-Based System for Automated Duty Management

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Team Member: **BHANU PRIYA** (2520090171)

2025–26 | Term III

Project Abstract

Overview & Problem Statement

⚠ The Problem

Manual duty allocation is **time-consuming, error-prone**, and leads to uneven workload distribution among faculty staff.

✓ Our Solution

A smart, web-based system that **automates and centralizes** invigilation duty assignments with fairness and transparency.

Key Features



Centralized Management

Faculty, schedules & rooms in one platform



Fair Distribution

Based on availability & workload balance



Notifications

Real-time alerts to faculty members



Report Generation

Duty charts for coordination & records

Project Objectives

01

Automate Duty Assignment

Eliminate manual, error-prone scheduling processes with intelligent automation.

02

Ensure Fairness & Balance

Distribute workload evenly based on faculty availability, department, and prior assignments.

03

Reduce Conflicts

Prevent scheduling clashes and overlapping duty allocations automatically.

04

Enhance Transparency

Provide real-time access for both administrators and faculty members.

05

Improve Efficiency

Save administrative time and effort through streamlined operations.

06

Centralized Records

Maintain organized duty charts, reports, and exam schedule data in one place.

Purpose & Problem Statement

Why this system is needed



Purpose

Design a smart, web-based system that automates the allocation of examination invigilation duties.

It ensures fair, conflict-free and transparent assignment of faculty across exam halls, while greatly reducing manual administrative effort.



Problem Statement

Invigilation duties are assigned manually using spreadsheets and paperwork.

This is slow, error-prone, and often leads to uneven workloads, scheduling clashes and last-minute confusion among faculty and staff.

Goals

Target outcomes of the system

01 Automation

Replace manual allocation with rule-based automatic duty assignment.

02 Fairness

Balance the invigilation load evenly across all eligible faculty.

03 Conflict-Free

Prevent overlaps, double-bookings and schedule clashes.

04 Transparency

Give admins and faculty real-time visibility of all duties.

05 Speed

Cut allocation time from hours of work down to minutes.

06 Centralization

Keep schedules, rooms and duty records in one single place.

Project Scope

What the system covers — and what it does not



In Scope

Faculty profile & availability management

Automated invigilation duty allocation

Exam scheduling & room / hall allocation

Conflict detection & resolution

Faculty dashboard & notifications

Duty charts & report generation



Out of Scope

Student exam registration & enrollment

Question paper setting or evaluation

Salary, payroll or HR management

Biometric / CCTV exam monitoring

Native mobile app (web-based only)

Integration with external university ERP

System Users

Who interacts with the system



Administrator

Admin

Manages faculty data, configures fairness rules, runs allocation, approves duties and generates reports.



Faculty / Invigilator

Faculty

Views assigned duties, dates, timings and hall details, gets notifications and checks duty history.



Exam Coordinator

Coordinator

Sets up exam schedules and room allocations, and monitors conflicts and overall duty status.

Functional Requirements

What the system must do

FR1 Faculty Management

Add, update and manage faculty profiles, departments and availability.

FR2 Duty Allocation

Auto-assign invigilation duties using availability and fairness rules.

FR3 Exam & Room Scheduling

Create exam timetables and allocate examination halls and rooms.

FR4 Conflict Detection

Identify and flag clashes, overlaps and duplicate assignments.

FR5 Notifications

Send real-time alerts to faculty for new or changed duties.

FR6 Reports & Dashboards

Generate duty charts and provide role-based dashboards.

Non-Functional Requirements

How well the system must perform



Performance

Allocate duties for a full exam schedule within a few seconds.



Usability

Clean, intuitive web interface usable with minimal training.



Security

Credential-based login with role-based access control.



Reliability

Consistent, accurate allocation with no loss of data.



Scalability

Support growing numbers of faculty, exams and rooms.



Availability

Accessible anytime through any modern web browser.

Assumptions & Acceptance Criteria

Conditions and measures of success



Assumptions

Faculty availability data is kept up to date.

Exam schedules and room lists are provided.

Users have a device with a modern browser.

A stable internet connection is available.

Faculty respond to notifications in good time.



Acceptance Criteria

Duties are allocated with no scheduling conflicts.

Workload is distributed fairly across faculty.

Faculty receive accurate, timely notifications.

Admin can generate correct duty charts & reports.

System runs reliably on standard web browsers.

Module Overview

4 Core Modules | 2 Team Members

01



Administration & Duty Allocation

Assigned To

LAHARI

2520090226

02



Exam Scheduling & Room Allocation

Assigned To

BHANU PRIYA

2520090171

03



Invigilation Duty Assignment

Assigned To

LAHARI

2520090226

04



Faculty Dashboard & Notification

Assigned To

BHANU PRIYA

2520090171

Administration & Duty Allocation Module

Assigned to: LAHARI (Roll No: 2520090226)

Core Responsibilities

Faculty Detail Management

Register, update, and manage complete faculty profiles including department and availability data.

Duty Assignment Control

Assign, reassign, and oversee invigilation duties with audit trail and approval workflow.

Admin Dashboard

Centralized view of all allocations, conflicts, and scheduling status in real-time.

System Configuration

Set rules for fair distribution, department quotas, and workload thresholds.

Exam Scheduling & Room Allocation Module

Assigned to: BHANU PRIYA (Roll No: 2520090171)

Core Responsibilities

Exam Schedule Management

Create and manage examination timetables including date, time, subject, and session details.

Room Allocation

Assign and optimize examination hall allocations to prevent conflicts and overcrowding.

Conflict Detection

Automatically detect and flag scheduling conflicts, duplicate room bookings or overlaps.

Schedule Records

Maintain organized records of past and upcoming exam schedules for reference.

Invigilation Duty Assignment Module

Assigned to: LAHARI (Roll No: 2520090226)

Duty Assignment Flow

1

Input Faculty Data

2

Check Availability

3

Apply Fairness Rules

4

Generate Assignments

5

Conflict Resolution

6

Final Allocation

Key Capabilities

✓ Auto-allocate based on workload

✓ Avoid duplicate assignments

✓ Department-wise distribution

✓ Override & manual adjustments

Faculty Dashboard & Notification Module

Assigned to: BHANU PRIYA (Roll No: 2520090171)

Faculty Dashboard — Preview

 Exam Date **12-June-2025**

 Reporting Time **8:30 AM**

 Room Number **Hall A-201**

 Subject **Frontend Dev**

 Assigned By **Admin**

 Status **Confirmed**

Notifications

Instant alerts sent to faculty for new or changed duty assignments.

Duty History

Faculty can view past and upcoming invigilation records anytime.

Secure Login

Credential-based authentication for safe access to personal duty info.

Report Access

Download or view duty charts and official invigilation reports.

Technology Stack

Frameworks, Platforms & Tools Used



Frontend

HTML5

CSS3

JavaScript



Backend

PHP

Node.js



Database

MySQL



Dev Tools

VS Code

Google Chrome

System Workflow

How the Examination Invigilation Duty Allocator Works



Admin Login

Administrator logs in to the system with secure credentials.



Setup Data

Add faculty profiles, exam schedules, and room details.



Auto-Allocate

System assigns duties based on rules, availability & fairness.



Review & Approve

Admin reviews assignments and resolves any conflicts.



Notify Faculty

Faculty receive notifications with full duty details.



Faculty Access

Faculty log in to view duties, dates, timings & hall info.

Team Details



Team Lead

LAHARI

Roll No: 2520090226

Modules Handled

- Administration & Duty Allocation Module
- Invigilation Duty Assignment Module



Team Member

BHANU PRIYA

Roll No: 2520090171

Modules Handled

- Exam Scheduling & Room Allocation Module
- Faculty Dashboard & Notification Module

Conclusion

The Examination Invigilation Duty Allocator is a modern, web-based solution that brings efficiency, fairness, and transparency to one of the most tedious administrative tasks in academic institutions.

✓ **Automates**

Manual duty allocation completely eliminated

✓ **Fair & Balanced**

Workload distributed equitably among faculty

✓ **Transparent**

Real-time visibility for both admin and staff

✓ **Efficient**

Saves time, reduces errors, improves coordination

✓ **Reliable**

Scalable solution for colleges and universities

Thank You!